# Student Performance Analysis

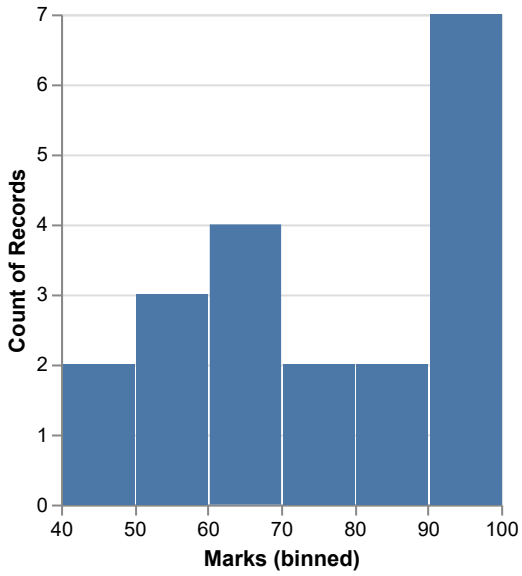
## Executive summary

This analysis covers **20 students**. Marks average **74.6** (median **75.5**) and range from **45 to 99**, indicating meaningful variation in achievement. Nine students scored in the **80–100** range, six scored **60–79**, and five scored below **60**.

Attendance and marks show a **moderate negative association** in this small sample (**r = −0.524, p = 0.018**). This should be interpreted as a pattern in the observed records—not evidence that reduced attendance improves marks.

## Marks distribution

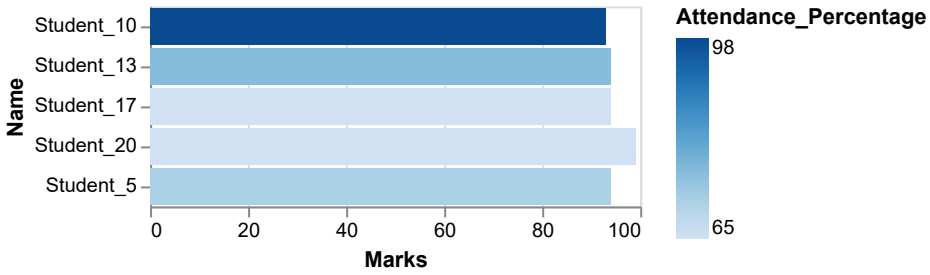
The marks are broadly dispersed rather than concentrated around a single score band.




| Performance band | Students | Share |
|---|---|---|
| 80–100 | 9 | 45% |
| 60–79 | 6 | 30% |
| Below 60 | 5 | 25% |

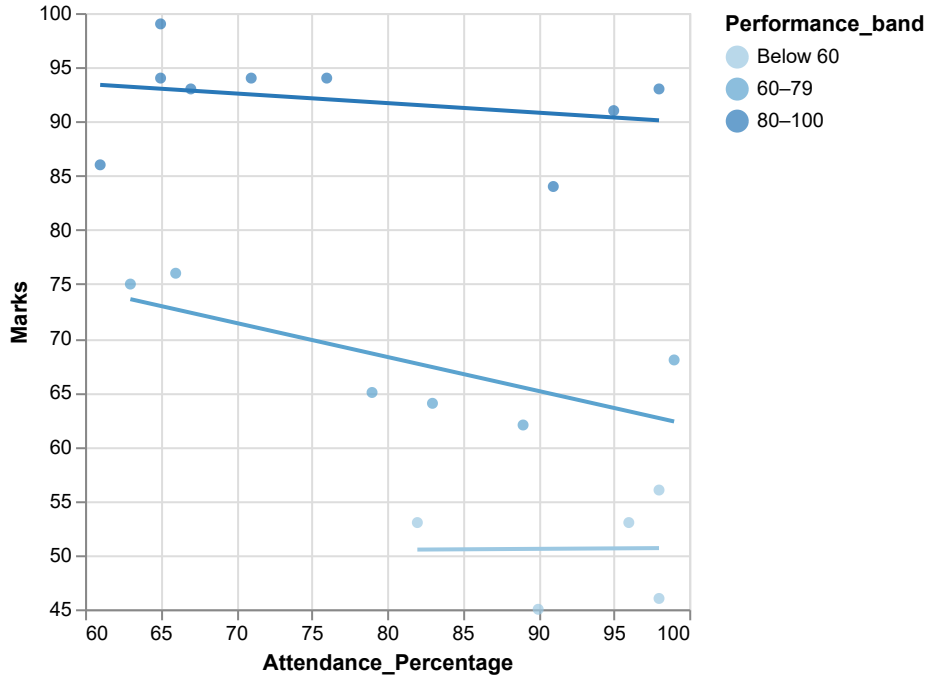
# Highest-scoring students

| Rank | Student | Marks | Attendance |
|---|---|---|---|
| 1 | Student_20 | 99 | 65% |
| 2 | Student_13 | 94 | 76% |
| 3 | Student_5 | 94 | 71% |
| 4 | Student_17 | 94 | 65% |
| 5 | Student_10 | 93 | 98% |




# Attendance and marks

Students with attendance below **80%** averaged **86.2** marks, versus **65.0** for students with attendance of **80% or more**. The regression view below shows the observed relationship and its substantial student-level variation.




# Interpretation and limitations

The lower-attendance group performs better on average in this particular dataset, but the dataset is small and may reflect unmeasured differences such as prior attainment, course difficulty, assessment timing, or data-entry issues.

Correlation does not establish causation. Attendance remains an important educational engagement measure and should not be treated as detrimental based on this sample.

Follow-up analysis should use a larger cohort and incorporate additional context—such as subject, assessment type, prior performance, and socioeconomic or support factors—before making intervention decisions.

# Takeaway

Achievement is mixed, with nearly half of students scoring 80 or above but one quarter below 60. The attendance pattern is noteworthy for investigation, yet it is not sufficient to support a causal conclusion or a change in attendance policy.